AP Precalculus **Unit 1, Lesson 1.10**

Exit Ticket

Name: ____________________________ Date: ________________ Period: _________

Consider the rational function $r(x) = \dfrac{x^2 - x - 2}{x^2 - 4}$.

1. Factor the numerator and denominator completely.

   Numerator: $x^2 - x - 2 =$ _____________________________

   Denominator: $x^2 - 4 =$ _____________________________

2. List any shared zeros of the numerator and denominator.

   Shared zero(s): _______________________

3. For each shared zero, compare multiplicities and classify as a **hole** or a **vertical asymptote**. Then state any additional VAs from non-shared zeros.

Zero	Num. mult. m	Denom. mult. n	Hole or VA?
_________	_______	_______	_________________
_________	_______	_______	_________________

4. Find the coordinates (c, L) of any holes. Show the simplification and substitution.

   Simplified form: _______________________

   $L =$ _______________________ Hole at: $\left(\_\_\_\_\_, \_\_\_\_\_\_\_\_\right)$

5. Write the limit statement for the hole using proper notation.

   $\lim\limits_{x \to \_\_\_\_\_\_} r(x) =$ ____________